



Python Programming Fundamentals

Sets and Frozensets

Programming Fundamentals Team

SETU

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Outline

1. What is a Set?
2. Creating Sets
3. Modifying Sets
4. Set Operations
5. Practical Examples
6. `frozenset`
7. When to Use Sets



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1.1 Unordered collection of unique values

```
fruits = {"apple", "banana", "cherry", "apple"}  
print(fruits)    # {'apple', 'banana', 'cherry'} ← apple deduped  
print(len(fruits))    # 3
```

Key properties:

- **No duplicates** — each value appears at most once
- **Unordered** — no index, no guaranteed order
- **Mutable** — can add/remove elements
- **Elements must be hashable** (strings, ints, tuples — but not lists)



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2. Creating Sets

```
# Literal syntax (note: {} alone creates empty dict!)
```

```
s = {1, 2, 3}
```

```
# From a list (deduplication!)
```

```
numbers = [1, 2, 2, 3, 3, 3, 4]
```

```
unique = set(numbers)
```

```
print(unique)    # {1, 2, 3, 4}
```

```
# Empty set
```

```
empty = set()    # NOT {} (that's a dict)
```

```
# From a string (unique characters)
```

```
chars = set("hello world")
```

```
print(chars)     # {'h', 'e', 'l', 'o', ' ', 'w', 'r', 'd'}
```



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3. Modifying Sets

```
s = {1, 2, 3}

s.add(4)          # {1, 2, 3, 4}
s.add(2)          # {1, 2, 3, 4} – no duplicate
s.remove(2)       # {1, 3, 4} – KeyError if missing
s.discard(99)     # {1, 3, 4} – safe, no error if missing

# Membership test – O(1)!
print(3 in s)     # True
print(99 in s)    # False
```




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4.1 Union, Intersection, Difference

```
A = {1, 2, 3, 4, 5}
```

```
B = {3, 4, 5, 6, 7}
```

```
# Union – all elements from both
```

```
print(A | B)           # {1, 2, 3, 4, 5, 6, 7}
```

```
print(A.union(B))      # same
```

```
# Intersection – only in both
```

```
print(A & B)           # {3, 4, 5}
```

```
print(A.intersection(B)) # same
```

```
# Difference – in A but not B
```

```
print(A - B)           # {1, 2}
```

```
print(A.difference(B))  # same
```



4.1 Union, Intersection, Difference

```
# Symmetric difference – in one but not both  
print(A ^ B)                # {1, 2, 6, 7}
```



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5. Practical Examples

```
# Find common students in two courses
python_students = {"Alice", "Bob", "Carol", "Dave"}
java_students    = {"Bob", "Eve", "Alice", "Frank"}

both = python_students & java_students      # {'Alice', 'Bob'}
only_python = python_students - java_students # {'Carol', 'Dave'}
all_students = python_students | java_students

print(f"In both courses: {both}")
print(f"Python only: {only_python}")
print(f"Total unique students: {len(all_students)}")

# Deduplication
product_ids = [101, 102, 101, 103, 102, 104]
unique_ids = set(product_ids)
print(unique_ids)      # {101, 102, 103, 104}
```



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6.1 Immutable set — can be used as a dict key

```
fs = frozenset([1, 2, 3])
print(fs)          # frozenset({1, 2, 3})

# frozenset can be a dict key (regular set cannot)
permissions = {
    frozenset(["read"]): "Reader",
    frozenset(["read", "write"]): "Editor",
    frozenset(["read", "write", "admin"]): "Admin",
}

user_perms = frozenset(["read", "write"])
role = permissions.get(user_perms, "Unknown")
print(role)        # Editor
```



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7. When to Use Sets

Use set when...	Don't use set when...
Deduplication needed	Order matters
Fast membership testing	Indexing needed
Set operations (union, etc.)	Duplicate values matter
Unique items only	Items are unhashable (lists)



Thanks for Watching - Any questions?

